

COMPLEX VARIABLE

BASIC PART: I

1) Complex Number:

$$z = x + iy \quad \text{or} \quad re^{i\theta}$$

2) Complex conjugate [i ka sign change]:

$$\bar{z} = x - iy \quad \text{or} \quad re^{-i\theta}$$

3) Modulus:

$$r = |z| = |x + iy| = \sqrt{x^2 + y^2}$$

4) Amplitude:

$$\theta = \tan^{-1} y/x$$

5) Function of 'z':

$$F(z) = u + iv = w$$

6) Euler's Formula:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

7) Principle Value :

$$\log(x + iy) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}(y/x) = \log r + i\theta$$

8) Compound Angle:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\sinh(A \pm B) = \sinh A \cosh B \pm \cosh A \sinh B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\cosh(A \pm B) = \cosh A \cosh B \pm \sinh A \sinh B$$

9) Imaginary Angle:

$$\sin ix = i \sinh x$$

$$\sinh ix = i \sin x$$

$$\cos ix = \cosh x$$

$$\cosh ix = \cos x$$

10) Derivative:

$$1) \quad \frac{d}{dx}(\text{const}) = 0$$

$$2) \quad \frac{d}{dx}(x^n) = n \cdot x^{n-1}$$

$$3) \quad \frac{d}{dx}(e^x) = e^x$$

$$4) \quad \frac{d}{dx} \frac{1}{x} = -\frac{1}{x^2}$$

$$5) \quad \frac{d}{dx} [\log f(x)] = \frac{1}{f(x)} \cdot \frac{d}{dx} [f(x)]$$

$$6) \quad \frac{d}{dx} [\tan^{-1} f(x)] = \frac{1}{1 + f(x)^2} \cdot \frac{d}{dx} [f(x)]$$

$$7) \quad \frac{d}{dx} [u \cdot v] = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$8) \quad \frac{d}{dx} (u/v) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

9) $ds = c$	$\int s = -c$
$dc = -s$	$\int c = s$

11) Integration:

$$1) \quad \int \text{const} \cdot dx = [\text{constant} \times x] + c$$

$$2) \quad \int x^n \cdot dx = \frac{x^{n+1}}{n+1} + c$$

$$3) \quad \int e^x \cdot dx = e^x + c$$

$$4) \quad \int (1/x) \cdot dx = \log x + c$$

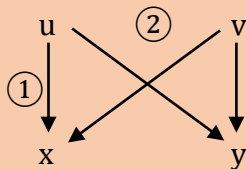
$$5) \quad \int \sec^2 x \cdot dx = \tan x + c$$

$$6) \quad \int \operatorname{cosec}^2 x \cdot dx = (-\cot x) + c$$

Derivative and integration of hyperbolic function is always positive

TYPES OF FUNCTIONS

Analytic Function



$$i) \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$ii) -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

Harmonic Function

u or v

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

DERIVATIVE

NORMAL

d

$$z = x^2 + y^2$$

$$\frac{dz}{dx} = 2x + 2y \left(\frac{dy}{dx} \right)$$

PARTIAL

∂/∂

$$z = x^2 + y^2$$

$$\frac{\partial z}{\partial x} = 2x + 0 = 2x$$

EXAMPLE ON POLAR FORM

$$z = re^{i\theta}$$

Analytic Function

$$i) \frac{\delta u}{\delta r} = \frac{1}{r} \frac{\delta v}{\delta \theta}$$

$$ii) +\frac{1}{r} \cdot \frac{\delta u}{\delta \theta} = -\frac{\delta v}{\delta r}$$

u
 $\downarrow$
 δr

v
 $\downarrow$
 $r\delta\theta$

Harmonic Function

$$\frac{\delta^2 u}{\delta r^2} + \frac{1}{r} \frac{\delta u}{\delta r} + \frac{1}{r^2} \cdot \frac{\delta^2 u}{\delta \theta^2} = 0$$

$$\frac{\delta^2 v}{\delta r^2} + \frac{1}{r} \frac{\delta v}{\delta r} + \frac{1}{r^2} \cdot \frac{\delta^2 v}{\delta \theta^2} = 0$$

STANDARD EXAMPLES

$$1. z \cdot \bar{z} = |z|^2 = |\bar{z}|^2$$

$$2. f(z) \cdot f(\bar{z}) = |f(z)|^2 = |f(\bar{z})|^2$$

$$3. f'(z) \cdot f'(\bar{z}) = |f'(z)|^2 = |f'(\bar{z})|^2$$

$$4. \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} = 4 \frac{\delta^2}{\delta z \cdot \delta \bar{z}}$$

